

ALLIANCE GIRLS 2025 PREDICTION



POSSIBLE KCSE EXAM

CHEMISTRY

233/3

(PRACTICAL)

PAPER 3

TIME: 2¼ HOURS

SCHOOL.....

SIGN.....

Kenya Certificate of Secondary Education.

CONFIDENTIAL

In addition to the fittings and apparatus found in chemistry laboratory, each student will require the following.

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| 1. 200cm ³ of hydrochloric acid, solution P . | 15. Six dry test-tubes. |
| 2. 150cm ³ of sodium hydroxide, solution Q . | 16. About 1cm by 3cm aluminium foil |
| 3. 0.50g of solid B weighed accurately. | 17. 1 red and 1 blue litmus papers. |
| 4. About 0.2g of solid Y . | 18. Universal indicator paper |
| 5. About 0.5g of solid M . | 19. About 2cm ³ ethanol placed in a test tube |
| 6. About 0.5g of solid W . | 20. 2 boiling tubes. |
| 7. Burette 0-50 cm ³ . | 21. Metallic spatula |
| 8. Pipette 25 cm ³ . | 22. Funnel. |
| 9. One pipette filler | 23. A test-tube holder. |
| 10. Means of labeling (one label). | 24. Stop clock (wall clock) |
| 11. 100 cm ³ measuring cylinder. | |
| 12. 10ml measuring cylinder | |
| 13. 250 cm ³ beaker. | |
| 14. Two conical flasks. | |

BENCH REAGENTS/ ACCESS REAGENTS

1. *Methyl orange indicator – supplied with dropper*
2. *Distilled water in a 500ml wash bottle*
3. *2M sodium hydroxide – supplied with dropper*
4. *2M aqueous ammonia – supplied with dropper*
5. *0.5M lead (II) nitrate solution – supplied with dropper*
6. *Source of heat*
7. *Universal indicator solution supplied with a dropper*
8. *pH chart*
9. *Acidified potassium manganate (VII) – supplied with a dropper*
10. *Bromine water – supplied with a dropper*

NOTES

1. *Solution **P** is prepared by adding 18.0cm³ (density=1.18g/cm³) of concentrated hydrochloric acid into 600cm³ of distilled water contained in a one litre volumetric flask and diluting to one litre of solution.*
2. *Solution **Q** is prepared by dissolving 8.80g of sodium hydroxide in 600cm³ of distilled water contained in a one litre volumetric flask and diluting to one litre of solution. (**0.22M**)*
3. *Acidified potassium manganate (VII) is prepared by dissolving 2.0g of potassium manganate (VII) provided in about 100cm³ of 2M sulphuric (VI) acid, adding 800cm³ of distilled water and diluting to one litre of solution. Label this as acidified potassium manganate (VII).*
4. *Bromine water is prepared by adding 2ml of liquid bromine to 100cm³ of distilled water and the mixture stirred well in a fume cupboard.*
5. *Solids :*
 - B is Na₂CO₃**
 - ◆ **M is Ba(NO₃)₂**
 - ◆ **W is Maleic acid**
 - ◆ **Y is NaHCO₃**